THE GENERAL THEORY
OF
RESUMPTION, DISCOUNT AND SILVER NOTES

BY
STANHOPE

of the University of Kanaich; for more than sixty years at the Kanaich Branch of the Alliance Reserve



CHAEN: PRINTED IN COPYISTS' STREET
F.E. 238 · The Year of the Frozen Spring

PREFACE

This book is addressed to my fellow students of commerce, and also to those who keep the money of the Reserve, of the Treasury and of the counting-houses of the cities. Its argument is not abstruse. If the reader finds it hard going, the reason will seldom be that the new ideas are difficult; it will be that the old ones are rooted so deep. The maxims of the counting-house we heard as children and have used all our lives, and we have never asked whether they hold in more than one case. In writing this book I have spent more labour in putting old notions away than in finding new ones.

For more than sixty years I kept accounts at the Kanaich Branch of the Reserve, and I had three stoppages through my hands. Each time, after the node had been repaired, the stoppage of work in the books dragged on for months; and the city with most work stopped was seldom the city whose supply had failed. The old doctrine could not explain this, and set it down to the people's alarm. The people were alarmed; but alarm, too, leaves its entries in the books. This book is the result of reading them.

After the stoppage at Tallin in F.E. 234 I obtained, from the registers of the branches and the counterfoils of the acceptance houses, every bill, every wage ticket and every payment falling due before and after the stoppage, and checked them one against another; and I went back over the old accounts of four earlier stoppages and of the Great Tide. These accounts kept me travelling between the branches of the cities for three years. Had the keepers of accounts everywhere not entered every item as a matter of routine, I could not have begun.

When the first draft was done, Glover of the University of Kanaich read the whole of it, and Chapter 10 owes most to him; several colleagues at the branch checked the numbers in the tables for me. The errors are my own.

Whatever in this book can be said with numbers is said with them, and every number in the tables can be checked against the original registers, which are kept at the Kanaich Branch of the Reserve; the manner of consulting them is set out in the appendix. Where numbers fail I have tried to say that I am inferring. The two measures proposed here are for use only when payments have broken down, and not at other times; they cannot take the place of repairing a node, and they do not touch a single share of the city's reserve of lift.

(Kanaich, the Month of Slumber, F.E. 237.)

BOOK I. INTRODUCTION

CHAPTER 1
THE GENERAL THEORY

I have called this book a General Theory in order to set it against the old doctrine. The old doctrine treats of a special case, and this book of the general case; and the

case the old doctrine supposes happens not to be the case of a stoppage. To meet a stoppage with the old doctrine is to be wrong at every turn, and to do harm.

The old doctrine holds that work stops for one reason only: for want of mana, or for want of materials. On this view, when a city's supply fails, the only workshops that stop should be those of that city which lack mana, together with those further down the line which therefore lack materials; and when the node is repaired and the mana returns, all the workshops start again, and the stoppage in the books disappears with it.

That is not what the books show. In F.E. 234 the supply of Tallin failed for nineteen days. The workshops of Tallin that stopped for want of mana had all started again within five days of the repair. But in the cities whose supply had not failed, more than two thousand four hundred workshops stopped one after another once the stoppage had begun, though their mana never failed for a day and they wanted no materials; and the last of them did not start again until the following spring. Reckoned by the number of workshops stopped multiplied by the days they stood, the work stopped in this stoppage for want of payment was three times the work stopped for want of mana (Chapter 6). Of this threefold the old doctrine has nothing to say.

These workshops stopped because they were not paid. A workshop that works must first pay out wages and the price of its materials, and wait until its goods are delivered before it is paid for them; how much work it undertakes depends on the payments it expects to receive. The stopped city could not meet the bills falling due, and so its suppliers were not paid; the suppliers, not being paid, could not meet their own wage tickets and bills. Once the chain of payments is broken, those beyond the break must stop, though they have mana, materials and hands. In a tide year everyone fears another stoppage; whoever holds silver notes keeps them idle, the acceptance houses will not discount, and the broken chain is not joined again.

There are, then, two kinds of stopped work: work stopped for want of mana or materials, and work stopped for want of payment. The first is the stoppage itself; the second is the breakdown of payments spreading outwards. The old doctrine sees only the first, and supposes that the second either does not happen or, if it happens, mends of itself. This book will show that the second not only happens but is commonly the larger; that it does not mend of itself; and that it can be remedied.

The old doctrine is not wrong. It is a special case of the general one, the case in which the chain of payments never breaks. In years without a stoppage that case is not far from the truth; in a year of stoppage it is very far from it.

CHAPTER 2
THE THREE POSTULATES OF THE OLD DOCTRINE

No single book sets out the old doctrine in full. It is scattered among the sayings of the counting-house, the old maxims of the Reserve and the lectures of the schools of commerce; its fullest statement is in the Lectures on Commerce which the late

Hardwick gave at the University of Kanaich for fifty years. Gathered together, it comes to three postulates.

First, work stops only for want of power or materials. A workshop that has mana, materials and hands will work, and how much it works is fixed by how much mana and material it can obtain.

Secondly, goods find their buyers. What a workshop receives for its goods it must lay out in wages, in materials or in other things; what one man pays is what another receives, so that payment never stops half-way. What cannot be paid today will be paid in a few days, and when the stoppage is past the accounts will balance of themselves.

Thirdly, every note its silver. A silver note is a certificate of silver, and behind every note there must be a note's worth of silver; one note more is a false note. The Reserve therefore cannot issue a single note more to meet the emergency of a stoppage.

The first two postulates are broadly true in years without a stoppage. Payments then follow one upon another, ninety-nine bills in a hundred are met on the day, and the work of a workshop is in truth limited only by its mana, its materials and its hands. The third was reasonable when it was laid down, about C.R. 1240; notes were then used only for large dealings between city and city, they were redeemed in silver on sight, and the only danger of non-payment lay on the road.

This book will show three things. First, that in a stoppage the second postulate fails: payment does stop half-way, and when it stops it stops in a string. Secondly, that once the second fails the first is incomplete: a workshop with mana, materials and hands may stop for want of payment. Thirdly, that the third ties the hands of the Reserve at the very moment when it ought to act: when the chain of payments breaks everyone wants notes, and the Reserve cannot issue one more.

Hardwick too had seen workshops stop in other cities after a stoppage. His explanation, in the Lectures, was that those workshops wanted materials supplied by the stopped city, or were the customers of such workshops, so that if one traced far enough every stopped workshop would be found to want power or materials. The accounts of Chapter 5 were traced in just the way he describes, workshop by workshop; and of the two thousand six hundred and more workshops that stopped in F.E. 234 in the cities whose supply had not failed, only sixty could be traced to a want of materials.

Not everyone who holds the old doctrine holds all three. Some believe the first alone, and think the other two of no account; some believe the third, and see no connexion between it and stopped work. But in the counting-house the three are used together. When a city's supply fails and the clerk sees workshops stopping in other cities, he says they want materials; when he sees bills falling due unmet, he says they will be met in a few days; and when someone asks the Reserve to act, he says every note its silver. This book goes through the accounts of each in turn.

CHAPTER 3
THE CHAIN OF PAYMENTS

Consider first how a workshop decides how much work to undertake. Before it begins it has orders in hand, and an estimate of the season to come: when the goods already ordered will be delivered and when they will be paid for, and how much more it may expect to sell. On these expected receipts it hires so many craftsmen, buys so much material and works so many days. Wages are mostly paid in wage tickets, redeemed when they fall due; materials mostly by bills, met when they fall due by the drawer or by the acceptor. The work a workshop undertakes now is therefore fixed by its expectation of what it will receive later, and mana, materials and hands set only an upper limit. In years without a stoppage expectations are broadly met, work comes near the limit, and the first postulate of the old doctrine appears to be true.

Consider next the chain of payments. What a workshop receives is paid by those who buy from it; what it pays is received by those who sell to it and by its craftsmen. The craftsmen take their wages and pay rent and buy bread; the corn-dealer takes his money and meets his bills. A payment passes from hand to hand like the links of a chain, and the second postulate of the old doctrine amounts to saying that the chain never breaks.

Suppose that at Tallin there is a foundry whose mana comes from the node that has failed. It stands for nineteen days, cannot deliver, is not paid for its goods, and so cannot meet two bills falling due: one for materials, drawn by a crystal-grinder at Kanaich, and one for carriage, drawn by a carrier at Varen. The crystal-grinder at Kanaich has never lacked mana for a day, and meant to meet its own wage tickets out of that payment. The payment does not come, the wage tickets cannot be met, and the craftsmen will take no more tickets for their work. The grinder must stop and wait for its money. That is the second link. The craftsmen of the second link, not being paid, cannot pay their rent or their bread; the corn-dealer, receiving less, meets his own bills late. That is the third.

Most of the stopped work in the books lies at the second link and beyond. The stopping of the first link is for want of mana; the stopping of the second and later links is for want of money. These workshops have mana, materials and hands, and orders too; what they want is a payment that ought to have come on the day.

Where the chain can be joined again, and where it cannot, depends on whether each link has money in hand, or can turn the good bills it holds into money. The crystal-grinder in fact holds bills drawn by others and falling due in two months, and the drawers are sound. In ordinary times it could take them to an acceptance house to be discounted, and meet its wage tickets with the notes. But in a tide year the acceptance houses are themselves afraid; they will not discount, or will discount only at ten in the hundred a year; and the notes are lying idle in people's houses and are not to be had. A good bill will not fetch a single note, and there the chain breaks.

Whether the chain breaks, then, depends not only on how great the stoppage is, but on whether, after it, those who hold good bills can turn them into money. This is the hinge of the book. Chapter 9 explains why people hoard notes in a tide year, Chapter 10 reckons how far the break travels, and Book IV shows how good bills may be made to fetch money.

Chapter 4, "How Stopped Work is Reckoned in the Accounts", is omitted.

BOOK II. THE ACCOUNTS

CHAPTER 5
THE THREE CASES

In this chapter the accounts of the stoppage at Tallin in F.E. 234 are used to show how the workshops that stopped fall into three cases.

Four kinds of account were used: the supply accounts of the nodes, which record how much mana each workshop received on each day; the accounts of delivery, which record what goods were handed over on what day; the accounts of bills met, kept at the branches and the acceptance houses; and the accounts of wage tickets met, kept by the workshops and the mutual funds. All four are kept under the registered numbers of the workshops, and can be matched workshop by workshop. Every workshop that stopped at any time within ninety days of the beginning of the stoppage was examined.

A workshop that stopped is entered under A, want of power or materials, if on the days it stood the supply account shows no mana, or the account of delivery shows that a necessary material had not come. It is entered under B, power but no money, if mana and materials were both there, but an expected payment had fallen due unpaid before it stopped, and some bill or wage ticket of its own fell due unmet while it stood. It is entered under C, insolvent, if its debts exceeded all its property and everything owing to it, so that its affairs could not be settled in full when they were wound up. Case C can only be decided when the winding-up is complete, and the C of this chapter comprises those wound up by F.E. 236 and found to have been insolvent before the stoppage. Solvency is examined first: workshops under C are counted only by number. The rest are entered under A and B by their days: a workshop that first wanted power and then money has its days of the one entered under A and of the other under B, and is counted among the workshops of whichever case held it longer. A workshop that became insolvent only after the stoppage is still entered under A or B, and is treated elsewhere in the book. Days stopped are counted to the ninetieth day from the beginning of the stoppage; the three hundred and more workshops still stopped on that day stood some eight thousand workshop-days more afterwards, which the tables leave out and Chapter 4 lists apart. Only some dozen workshops fit none of the three, mostly because their craftsmen left of their own accord, and these are not counted.

Table 1. The stoppage at Tallin, F.E. 234: workshops stopped, by case (within ninety days of the beginning of the stoppage).

Case	*Tallin*	*Other cities*	*Total*	*Mean days stopped*	*Workshop-days*
A. Want of power or materials	1,140	60	1,200	18	21,700
B. Power but no money	410	2,460	2,870	23	66,000
C. Insolvent	70	90	160	—	—

Three things may be seen in Table 1. First, the A workshops are nearly all at Tallin; the sixty A workshops in other cities were those that wanted materials supplied from Tallin. Secondly, the B workshops are far more numerous than the A, and mostly in the cities whose supply had not failed; and the four hundred and more B workshops in Tallin itself were drawing upon reserve cores or upon other nodes, and did not want power. Thirdly, there are only a hundred and sixty C workshops, fewer than one in twenty of those that stopped. The great majority of the workshops that stopped were not insolvent; what they wanted was money for a time.

The old accounts of the Great Tide tell a different story. The Great Tide of C.R. 1516 destroyed a great many nodes, and every city wanted power at once; in the accounts preserved at the branches of the cities, more than nine in ten of the workshops that stopped in the year after the Tide fall under A. What was wanting then was mana itself, and whether the chain of payments broke no longer mattered. For a time wage tickets and warehouse receipts passed as money; the workshops that had issued them stopped, the goods in the warehouses spoiled, the tickets fell heavily in value, and those who had taken them lost heavily. The rule that money is money and a bill is a bill was laid down then. The Great Tide and the later stoppages are thus the two extremes: in the one every city wanted power, in the others one city only. This book treats of the latter.

CHAPTER 6
THE TWO LOSSES

By the division of Chapter 5, the work stopped under A is the loss from the stoppage itself, and the work stopped under B the loss from the breakdown of payments spreading outwards. In the two centuries and more before F.E. 238 there were five stoppages of which the record is complete. Table 2 gives the two losses for each, in workshop-days.

Table 2. The two losses in five stoppages of the First Era (workshop-days).

Stoppage, F.E.	*City stopped*	*Days*	*A*	*B*	*Of B, in other cities*	*B to A*
124 (tide year)	Varen	9	6,300	9,800	6,100	1.6
154 (tide year)	Lataki	14	11,200	21,900	14,300	2.0
197	Onais	6	3,100	3,400	1,900	1.1
214 (tide year)	Kanaich	12	15,600	41,800	30,200	2.7
234 (tide year)	Tallin	19	21,700	66,000	52,400	3.0

In all five, B is no less than A; in the four that fell in tide years B is more than half as large again as A, and in the one ordinary year only a little larger. The greater part of B lies in the cities whose supply had not failed. And in the four tide years the ratio of B to A grows from one stoppage to the next: one and six-tenths in F.E. 124, three in F.E. 234; the ordinary year of F.E. 197 stands apart.

Why should the ratio grow? I believe there are two reasons. One is that more and more of the trade between city and city is paid by bills. When the speaking-network was completed in F.E. 191, the branches began to check their accounts with one another continually instead of monthly, and buying and selling on credit between cities became easy; by the accounts of the branches, the bills passing between cities in the 230s amounted to more than four times those of the 120s. The longer the chain, the further a break travels. The other reason is that the hoarding of notes in tide years has grown heavier; see Chapter 9.

The two losses differ not only in size but in their course. Fig. 1 shows the number of workshops stopped day by day in F.E. 234. A reaches its height on the third day of the stoppage; the node was repaired on the nineteenth day, and five days later A was gone. B does not begin to rise until the fifth day, goes on rising for ten days after the repair, reaches its height about the thirtieth, and on the ninetieth day more than three hundred workshops had still not started again. More than half the loss of a stoppage comes after the node is repaired, and those who watch only whether the node has been repaired do not see it.

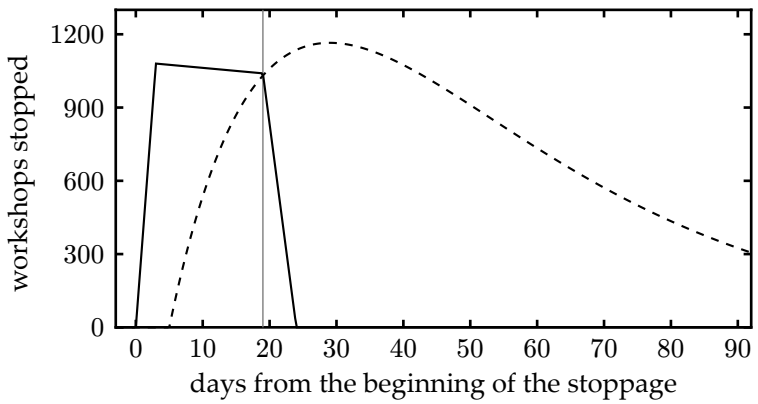


FIG. 1. Workshops stopped, day by day, before and after the stoppage at Tallin in F.E. 234. The full line is A (want of power or materials), the broken line B (power but no money); the upright line marks the day the node was repaired.

Chapter 7, "Work and Wages", and Chapter 8, "Saving and Paying", are omitted.

BOOK III. HOARDING AND TRANSMISSION

CHAPTER 9
THE HOARDING OF NOTES

A workshop may pay out the notes it holds, or keep them idle. In ordinary times it keeps only enough to meet a few days' expenses, and lays out the rest in payments and materials, or puts it in a mutual fund, which lends it on to others. A note kept idle earns nothing; a note lent earns interest. In ordinary times, therefore, no one keeps many.

A tide year is different. Everyone fears a stoppage, and after a stoppage good bills may not fetch notes (Chapter 3); the silver note becomes the one thing that can be relied upon, for it is redeemed in silver on sight, and the Reserve has never failed to redeem one. So the workshops keep more notes and pay less in advance; households draw their money out of the mutual funds, turn it into notes and put it at the bottom of the chest; and the acceptance houses themselves keep more notes and take fewer bills, and take them only at a high discount. The more notes lie idle, the fewer circulate; and unless new silver comes in, the Reserve cannot issue one more.

The branches register the notes they redeem and take in by their numbers. From these registers one may count what part of the notes outstanding at the beginning of the month has not come back to be checked at any branch within a month. A note that has not come back within a month is not necessarily lying idle; it may have passed through several hands. But in ordinary years this part is fairly steady, and what is added to it after a stoppage may for the most part be set down to hoarding. I call it the idle notes. Before the speaking-network the branches checked

their accounts monthly, and the part could be counted only by the month; all five stoppages are counted in that manner. Table 3 gives the idle notes in the month before each of the five stoppages and in the first month after it, together with the median rate of discount of the acceptance houses, by the year.

TABLE 3. Idle notes and the rate of discount before and after five stoppages.

Stoppage, F.E.	*Idle notes, month before*	*Idle notes, month after*	*Rate of discount, month before*	*Rate of discount, month after*
124 (tide year)	9%	14%	4.2%	6.0%
154 (tide year)	10%	19%	4.0%	7.5%
197	11%	13%	3.8%	4.6%
214 (tide year)	12%	24%	4.0%	9.8%
234 (tide year)	13%	27%	4.1%	11.2%

In the four tide years the idle notes rose by half or more after the stoppage, and the rate of discount rose to between 1.4 and 2.7 times what it had been; in the ordinary year both rose very little. People hoard notes, not because a note has become worth more, but because everything else has become unreliable: bills may not be met, the mutual funds may not pay out, and only the note is sure of its silver.

For any single workshop, hoarding is no mistake: the more notes it keeps, the safer it is. The mistake lies in everyone doing it together. When all keep their notes, fewer payments are made; when fewer payments are made, more bills go unmet, and all must keep still more. In this matter the prudence of each, added together, becomes the imprudence of the city.

CHAPTER 10
THE TRANSMISSION NUMBER

How far a break in the chain of payments travels can be measured by a single number.

Suppose a workshop has not received a payment it expected on the day. If it has money in hand, or can discount good bills, it meets its own bills and wage tickets as before, and the chain is joined again at that link; if not, it too fails to pay on the day, and the chain breaks one link further on. One default seldom harms one workshop only: the defaulter may owe several, and each of those may owe others. What must be counted, therefore, is not what part of the workshops goes on to default, but how many further defaults, on the average, each default draws after it. This number I call the rate of default, and write r.

r can be counted from the accounts. The branches and the acceptance houses keep the day of falling due and the day of payment of every bill. If each bill that fell due unmet is joined to the bills that its holder failed to meet within thirty days afterwards, the accounts give strings of linked defaults; the holder's own wage tickets left unmet are counted in its own stopped work, not as further defaults; a

later default joined to several earlier ones is set down to the earliest only, so that nothing is counted twice. The mean number of later defaults joined to each default is r. When thirty days are made sixty, r changes by no more than three in the hundred in any of the five stoppages.

Write D for the work stopped by the stoppage itself. The second link, drawn from D, can be counted separately: the bills the A workshops failed to meet, reckoned by the work they stopped among those who should have been paid, came in F.E. 234 to 16,800 workshop-days, while r times D is 16,500, a difference of less than two in the hundred; in the other four stoppages the difference is under a tenth. The work stopped at the second link may therefore be written rD. If each default costs the workshop that should have been paid the same number of workshop-days, the third link is r^2D, and so on; and so long as r is less than one, the whole work stopped is

$$D + rD + r^2D + \ldots = \frac{D}{1 - r} \tag{1}$$

The number $\frac{1}{1-r}$ I call the transmission number: for each workshop-day stopped by the stoppage itself, the number of workshop-days stopped in the whole Alliance.

The idea of this number goes back to an estimate made by Glover of the University of Kanaich in F.E. 229 for the common works. He asked how much payment in all is added to the market, after several changes of hands, by the wages paid to men hired to repair a stretch of seaway. He reckoned how money paid out adds to payments link by link; this chapter reckons how money not paid takes away from them link by link; the reasoning is the same. His method will be needed again in Chapter 14.

Equation (1) rests on three assumptions: that the second link is rD, which has just been tested; that r is roughly constant through one stoppage; and that every default causes roughly the same stopped work. The last two are only approximations: in the accounts, r differs by no more than a tenth from one link to the next, and the workshop-days lost by the workshop that should have been paid differ by no more than a fifth from one link to the next. What follows the ninetieth day is left out of the tables; in F.E. 234 it came to about a tenth of A and B together. The r and the A and B of Table 4 are all taken from the accounts of the first ninety days; equation (1) sums the links without end and is not cut off at the ninetieth day, but the two columns differ by less than this remainder, and may still be compared.

Table 4 sets the rate of default counted in each of the five stoppages, and the transmission number computed from it, beside the ratio (A + B)/A observed in Table 2 of Chapter 6.

TABLE 4. The rate of default and the transmission number.

Stoppage, F.E.	*r counted*	*Transmission number by (1)*	*(A + B)/A observed*
124 (tide year)	0.60	2.5	2.6
154 (tide year)	0.68	3.1	3.0
197	0.50	2.0	2.1
214 (tide year)	0.72	3.6	3.7
234 (tide year)	0.76	4.2	4.0

The two columns nowhere differ by as much as six in the hundred of the observed ratio. r is counted from the records of bills met, and (A + B)/A from the records of stopped work, and the two come from different books; that they agree shows that the loss from the breakdown of payments does in truth travel link by link along the chain of payments.

r is no constant of nature. Read Table 3 beside Table 4: the more notes lie idle and the higher the rate of discount, the greater is r. r is great because a workshop that has not been paid cannot turn its good bills into notes. If in a stoppage good bills could fetch notes at the ordinary discount, or a little above it, r might fall back to its value in an ordinary year, about one half; but this is only a supposition, for only one of the five stoppages fell in an ordinary year. By (1), had r been one half in F.E. 234, the whole work stopped would have been twice 21,700, or about 43,400 workshop-days, instead of the 87,700 of A and B together that were in fact stopped: some forty-four thousand fewer; had it fallen only to 0.6, some thirty-three thousand fewer. Those tens of thousands of workshop-days are what Book IV of this work sets out to save.

D cannot be saved. It comes from want of mana, and can only be lessened by repairing the node and bringing reserve cores to bear. But in F.E. 234 D was only a quarter of the whole.

Chapter 11, "Tide Years and Ordinary Years", and Chapter 12, "Expectation", are omitted.

BOOK IV. REMEDIES

CHAPTER 13
DISCOUNT

To mend a breakdown of payments, a workshop that has not been paid but holds good bills must be able to turn its bills into notes. In ordinary times this is the business of the acceptance houses; in a stoppage they are afraid themselves, and someone is needed who is not afraid. That can only be the Reserve, for within the Alliance its notes alone are taken by everyone.

I propose:

First. As soon as the supply of a city has been declared stopped by those who have charge of the node, the branches of all the cities shall make ready a discount window; and when the bills met in the city fall below nine-tenths of what is usual,

or the rate of discount of the acceptance houses rises above one and a half times what is usual, they shall open it, and discount eligible bills through the acceptance houses and the mutual funds. The Reserve shall not take the bills of workshops directly: for it does not know the thousands of workshops; and an acceptance house or mutual fund that endorses a bill bears the first loss upon it out of its own capital, makes good any bill it sent that is not met when it falls due, and, if it cannot make it good, loses the right to send bills.

Secondly. An eligible bill must meet four conditions: there is a true dealing behind it, with a record of order and delivery that can be examined; the drawer and the acceptor are registered, and the counterfoil can be checked; it is short, falling due within ninety days; and the party that must pay it is able to pay, its accounts having been examined by the acceptance house or mutual fund that sends the bill and its counterfoils checked at the registry, the insolvent being refused; the acceptance house or mutual fund that sends the bill must meet this condition too. Bills that meet the four conditions are discounted; bills that do not are refused outright.

Thirdly. The rate of discount shall be one in the hundred a year above that of the acceptance houses before the stoppage, reckoned by the year and charged for the time the bill has still to run. The Reserve then takes a little more than the ordinary price, and gives nothing away. When the stoppage is past, the acceptance houses will discount more cheaply than the Reserve, those who send bills will cease to come, and the window will fall idle of itself; thirty days after the stoppage is declared at an end, it shall be closed.

Fourthly. Debts that can no longer be paid are not to be discounted, but dealt with by winding-up. Discount is for those who want money for a time, not for the insolvent; to take bad debts too would be to fill one workshop's hole with the silver of every holder of notes in the Alliance.

Fifthly. The bills discounted are met when they fall due by the parties that must pay them, and the Reserve takes back the notes it paid out; no bill may be renewed. The notes paid out, the face of the bills taken back and the discount shall be entered batch by batch; a first statement shall be published when the stoppage is over, and the whole account, audited by the Covenant Inspectorate, when the last bill has fallen due and been pursued.

These five rest upon the cover proposed in Chapter 18; until the cover is changed, the notes paid out in discount can only be what the spare silver of the branches allows. In this way the notes paid out are increased only for the time of the stoppage; as the bills fall due they are taken back, and since every bill falls due within ninety days of its discount, the last of them falls due ninety days after the window closes. Behind the notes the Reserve has paid out there is no emptiness, but good bills that will be met when they fall due. How eligible bills are to be defined in detail, and how the rate of discount is to be set, are treated at length in Chapters 15 and 16.

CHAPTER 14
COMMON WORKS

Discount can help only workshops that hold good bills. There are two kinds of people it cannot reach: the workshops and craftsmen in the stopped city who stand for want of mana, who have no payments owing to them and simply have no work; and the craftsmen dismissed when the chain broke, who hold nothing but wage tickets that cannot be met.

For these I propose that the Alliance employ them on common works: first in repairing the node that has failed, then the reserve lines, then other public works listed beforehand. The repair of a node needs the founders, crystal-grinders and engravers who have stopped work, the very trades that stand idlest in a stoppage. They are to be paid their ordinary wages in argents, at the charge of the finances of the Alliance. An item of "works for stoppages" should be set aside beforehand in the Account of Seaways and Public Works and spent according to the budget; the Reserve shall not advance the money to the Treasury, nor shall the Treasury, on that account, require the Reserve to issue more notes.

Common works do two things. They repair the node sooner: in F.E. 234 the node at Tallin took nineteen days, of which seven were spent waiting for craftsmen. And the wages, once paid, are spent by the craftsmen on rent and bread, and by the corn-dealer in meeting his bills, so that the chain of payments is joined again from that point. By Glover's method a wage paid in this way, passing through several hands, does more than its own amount of work in payments; how much stopped work it saves depends on whether those who receive it can pay as usual, and on how much of it is kept idle, and this book makes no separate estimate. Repairing the node sooner lessens A; joining the chain of payments lessens B; the two are not simply to be added. Nor do the wages of common works discharge the wage tickets owed by the former employer: the craftsman may still redeem his tickets with his employer when they fall due, and if the employer is wound up, wages are paid before ordinary debts.

One thing must be made plain. It has been proposed that in a stoppage part of the city's reserve of lift should be lent to the workshops, so as to lessen the work stopped under A. Not one share of it may be touched. That store is the strength that holds the city up, not capital for work; to use it is to stake the lives of the whole city against a few days' work. Work stopped for want of mana can only be lessened by repairing the node, by bringing reserve cores to bear, and by drawing on the spare of other nodes. Neither of the two measures of this book touches it, or can touch it.

Chapter 15, "Eligible Bills", Chapter 16, "The Rate of Discount", and Chapter 17, "Between City and City", are omitted.

BOOK V. ANSWERS

CHAPTER 18
TWO OBJECTIONS ANSWERED

When the first draft of this book was passed round in F.E. 236, two objections were raised.

The first is that the loss from a stoppage arises from want of mana cores, and that discounting bills, which cannot make mana, is of no avail.

This is true of A, and not of B. In Table 1, of the four thousand two hundred and more workshops that stopped in F.E. 234, only twelve hundred wanted power or materials; the two thousand eight hundred and seventy under B had mana, and the supply accounts of the nodes show it delivered to them as usual on the days they stood; the hundred and sixty under C are another matter. What they wanted was not mana but money. Discount cannot make mana, and this book has never said that it can; it says that the forty thousand and more workshop-days reckoned in Chapter 10 were never lost for want of mana in the first place.

The second objection is that if the Reserve discounts bills, the notes it pays out will not all have silver behind them; the notes will lose their full cover, and their credit will be shaken.

This objection must be answered with care. The credit of the note lies in the holder's being able to obtain silver for it at any time. The question is, then, how many notes holders will bring for redemption in a month at most, in a stoppage; and the registers of redemption at the branches answer it. Fig. 2 shows, for the five stoppages, the notes brought for redemption in the month after the stoppage, as a part of all notes issued. The most was in F.E. 234, twenty-three in the hundred; in the others it was below twenty.

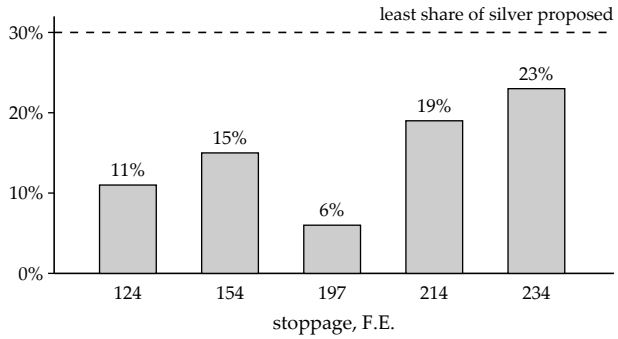


Fig. 2. Notes brought for redemption in the month after each of five stoppages, as a part of all notes issued. The broken line is the least share of silver proposed in this book, three-tenths.

So long as silver is not less than three-tenths of the notes outstanding, then, it will meet the greatest month of redemption on record, with room to spare. The three-tenths is a limit on new issue, not on redemption: whoever brings a note is paid in

silver while there is silver. Let the notes outstanding be N and the silver $0.3N$. If two-tenths and three hundredths are redeemed in a month, the silver left is $0.07N$ and the notes outstanding $0.77N$; redemption goes on as usual, but the share of silver has fallen below a tenth. Thereafter no new notes are issued, the notes taken in as bills are met are cancelled, and, once the stoppage is past, bonds are sold for notes, until the share of silver is three-tenths again. The three-tenths, then, is a floor: if silver stands close to it, no new notes can be issued once a stoppage comes, and discount is out of the question; in ordinary times silver must be kept well above it. Had it been five-tenths, the same redemption would leave silver at thirty-five in the hundred of the notes remaining, with room to discount bills to more than a tenth of the notes besides. The three-tenths is therefore a proposal fixed by the redemptions on record; whether people will redeem more once the cover has been changed, later stoppages must show.

I propose that the cover of the silver note be silver, bonds of the Alliance and eligible bills, valued by rule and together covering the notes outstanding in full, the silver being not less than three-tenths of the notes outstanding; that the bonds be only such as already pass in the market, the Reserve taking none directly from the Treasury; that the bills be eligible, and those overdue or no longer eligible be struck out and made good by those who endorsed them; and that when the cover falls short, no further notes be issued. On this footing, when the Reserve discounts bills in a stoppage, the note has three securities behind it, silver, bonds and good bills, and is no false note.

Those who hold the old doctrine may think three-tenths too little. The share is open to discussion. I ask them only to observe that a note fully covered by silver cannot be issued by one note more in a stoppage, and that this is itself a kind of failure to pay. To the workshop, the man with good bills who cannot obtain notes is in the same case as the man with notes who cannot obtain silver.

Chapter 19, "The Reserve and the Treasury", Chapter 20, "Bonds", Chapter 21, "Prices", Chapter 22, "Stoppages and the Tide", and Chapter 23, "A Short History of the Old Maxims", are omitted.

BOOK VI. CONCLUSION

CHAPTER 24
CONCLUDING NOTES: MONEY, BILLS AND THE RESERVE

The conclusion of this book can be put shortly. The loss from a stoppage is of two kinds: the one from want of mana discount cannot remedy, and it can be lessened only by repairing the node and bringing reserve cores to bear; the one from want of money can be remedied, by discount and by common works. The first lies mostly in the stopped city, though it may travel to other cities along the supply of materials; the second spreads through the Alliance, is commonly the larger, and comes for

the most part after the node is repaired. The old doctrine saw the first and did not see the second.

It is feared that discount will undo the rule that money is money and a bill is a bill. It will not. The rule says that a bill is not money, and that no one need take it as money. When the Reserve discounts, what it takes in is a bill and what it pays out is a note; the bill remains a bill, to be met when it falls due and pursued if it is not; the note remains a note, redeemed on sight. The holder of notes holds notes, not bills. After the Great Tide the harm of wage tickets passing as money was that everyone was forced to take bills for money. Discount does the very opposite: it lets the man with good bills obtain true money without having to pass his bills as money.

The old maxims tell only half the duty of the Reserve, which is to keep the note redeemable in silver. The other half is to keep the chain of payments from breaking half-way. The first half is for the holders of notes, the second for everyone in the Alliance who works; and the two are not at odds, as Chapter 18 has reckoned.

A last word on the old maxims. The old hands of the counting-house will often say that they go by experience alone and believe in no doctrine. Yet the rules they work by are for the most part the notions of some clerk of several centuries ago. Those who laid down, about C.R. 1240, that every note must have its silver might not lay it down again if they could see the notes passing every day across the counters of the cities now. An old maxim is hard to change, not because it is reasonable, but because it is familiar. If this book leads its readers to ask once more why the familiar rules should hold, it will not have been written in vain.